

the process. By adding a service worker and configuring it, you can transform your React app into a PWA.

- **Vue.js:** Vue.js provides plugin such as Vue CLI PWA plugin that facilitate the creation of PWAs. These plugins handle the setup and configuration, allowing you to focus on building your application.
- **Workbox:** Workbox is a set of libraries and node modules that make it easy to add offline support to your web apps. It simplifies service worker creation and management, enabling efficient caching and background sync.

Best practices for building high-performance PWAs

To ensure your PWA delivers an optimal user experience, follow these best practices.

- **Responsive design:** Design your PWA to work seamlessly across different devices and screen sizes. Use flexible grids, layouts, and media queries to ensure a consistent experience.

- **Efficient caching:** Utilise service workers to cache essential resources, enabling offline functionality and faster load times. Implement cache strategies that balance freshness and performance.
- **Fast load times:** Optimise your PWA for speed by minimising JavaScript, CSS, and HTML files. Use techniques such as lazy loading and code splitting to reduce initial load times.
- **Secure your app:** PWAs must be served over HTTPS to ensure data security and integrity. Implement HTTPS and use security headers to protect your application from threats.
- **Engaging user experience:** Leverage features like push notifications and home screen installation to keep users engaged. Use web app manifests to define your PWA's appearance and behaviour when installed.
- **Progressive enhancement:** Build

your PWA using progressive enhancement principles. Ensure your app works on all browsers and devices, providing basic functionality to older browsers while offering enhanced features to modern ones.

By embracing PWAs and following best practices, you can create applications that stand out in the crowded digital landscape, delivering a seamless and captivating experience to users.

Open source technologies offer powerful frameworks, tools, and best practices for building high-performance, secure, and scalable applications. From React Native for mobile to Django and Node.js for web, these solutions streamline development while fostering innovation and community collaboration. By prioritising user-centred design, performance optimisation, and robust security measures, developers can ensure their apps meet modern standards. **END** 

By: Meghraj Singh Beniwal

The author is the founder of Schooling for Next Generation, a tech enthusiast and Android hobbyist. He is a test automation analyst working in Virginia in the US.

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- **Results:** A significant boost in free-to-paid conversion rates, increased user engagement (including average listening hours), and a rise in desktop users.
- **Key features:** Lightweight design, offline functionality, and personalised playlists.

4. Telegram: From laggy app to agile PWA

- **Challenge:** Provide a smoother and faster user experience for sending messages and interacting with the platform.
- **Solution:** Replacing the outdated native app with a PWA.
- **Benefits:** Enhanced user experience

with faster performance, improved agility, and some offline functionality. This shift is likely a major contributor to Telegram's massive user base.

5. Facebook: A PWA for all devices

- **Challenge:** Offer a compelling alternative to the native app, especially for mobile users.
- **Solution:** A PWA that delivers a quick and efficient experience,

taking up minimal storage space.

- **Benefits:** Appeals to casual users and those with limited storage, while also offering a performant desktop experience.

The best part? PWAs are on the rise. From e-commerce giants to news outlets, businesses are recognising their potential. Don't get left behind! The future of mobile is fast, frictionless, and accessible. Make sure your business is part of it – go PWA today! **END** 

By: Dhaval Gajjar

The author is the CTO of Textdrip, and the CEO of Pranshtech Solutions and WeTechnolabs Solutions. With a passion for technology and innovation, he explores the ever-evolving world of digital solutions, sharing insights and expertise to drive progress in the tech industry.